

GSVA for mutil Group

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Contents

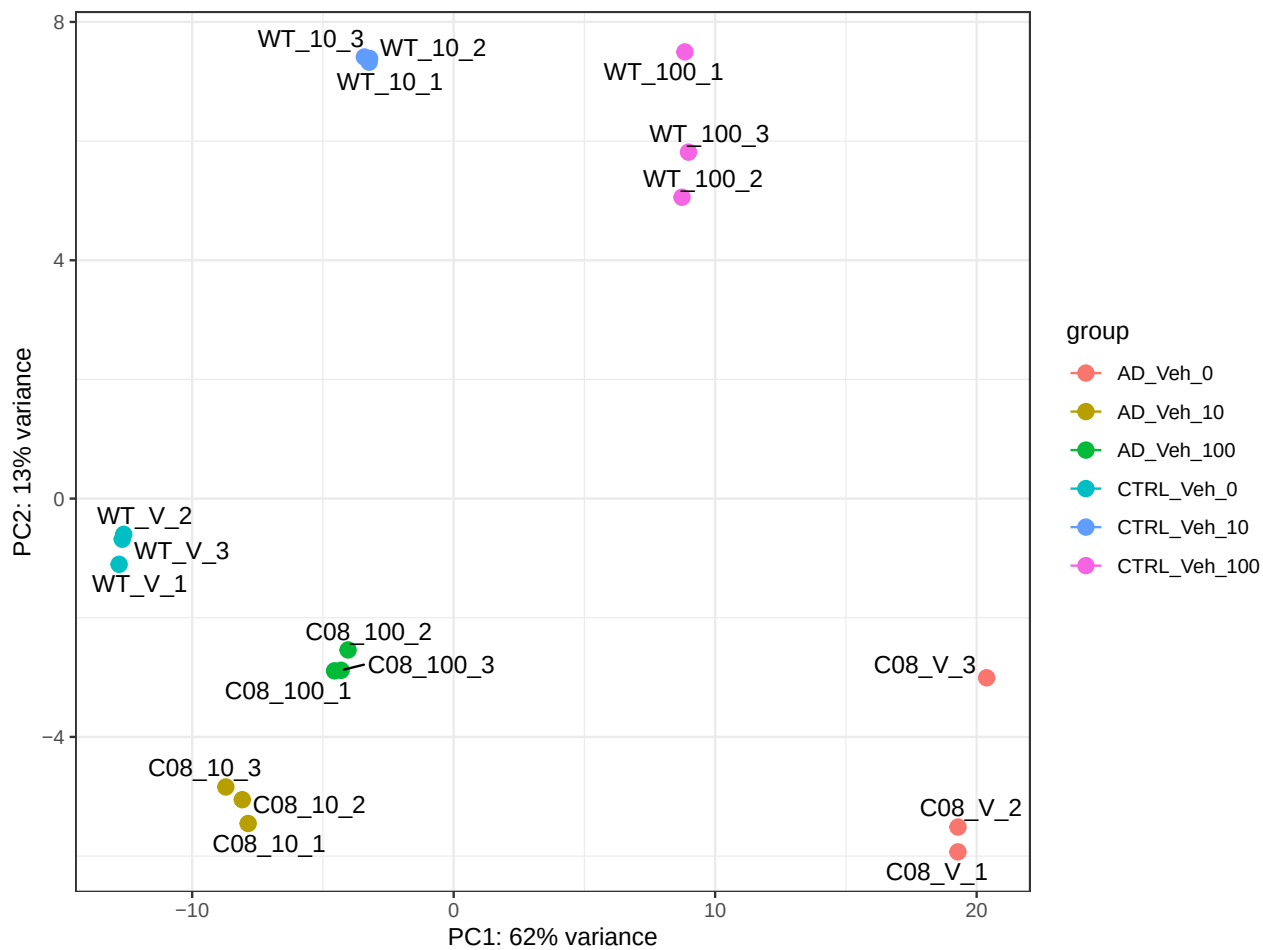
1. Read the count data	2
2. Visualization for reuslt	3
(1) Sample information - PCA plot	3
(2) Sample information - Distance heatmap	4
3. GSEA analysis and BioDomain analysis	5
(1) GSEA analysis for AD vs CTRL	5
(2) GSEA analysis for AD_Drug vs CTRL	5
(3) GSEA analysis for CTRL_Drug vs CTRL	5
4. BioDomain analysis	6
5. Load the GSVA analysis	6
(1) Ridge plot for GSVA score	6
(3) Rug and density plot for GSVA score	8

1. Read the count data

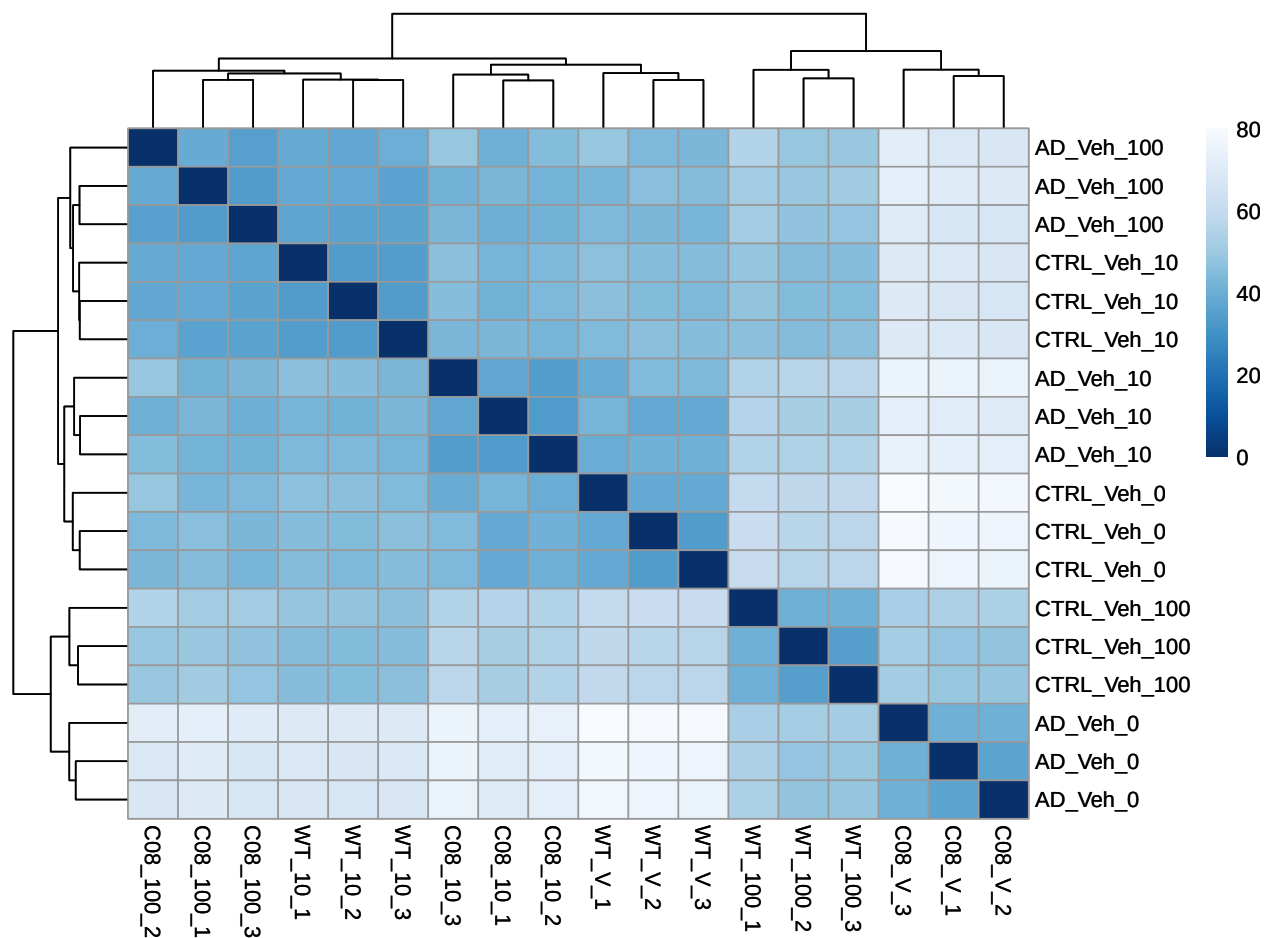
In this section, we will read the clean count data from the `synaptosomes_bulkRNA` folder. We will read the data and merge them into a single table.

2. Visualization for reuslt

(1) Sample information - PCA plot



(2) Sample information - Distance heatmap



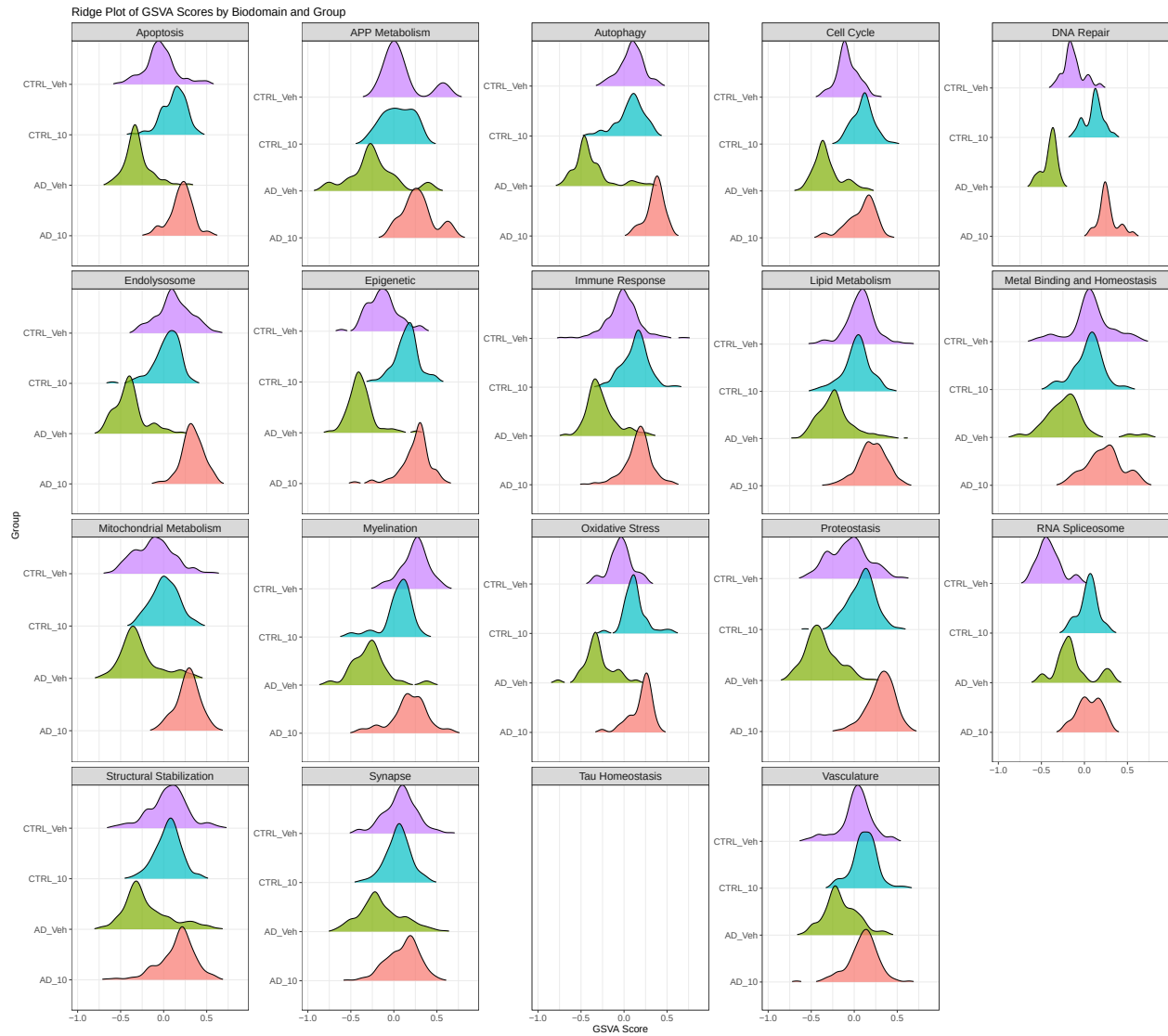
3. GSEA analysis and BioDomain analysis

- (1) GSEA analysis for AD vs CTRL
- (2) GSEA analysis for AD_Drug vs CTRL
- (3) GSEA analysis for CTRL_Drug vs CTRL

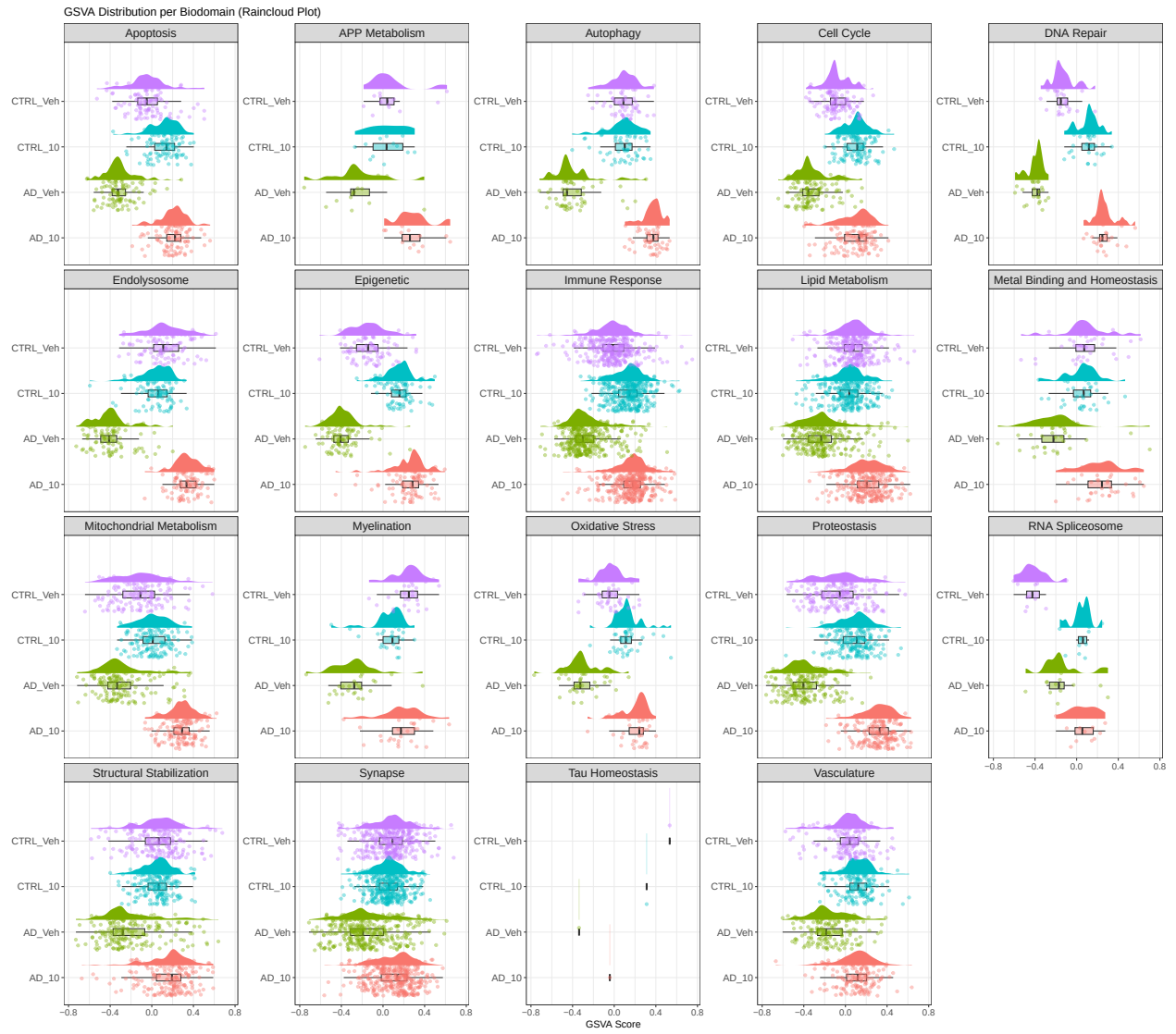
4. BioDomain analysis

5. Load the GSVA analysis

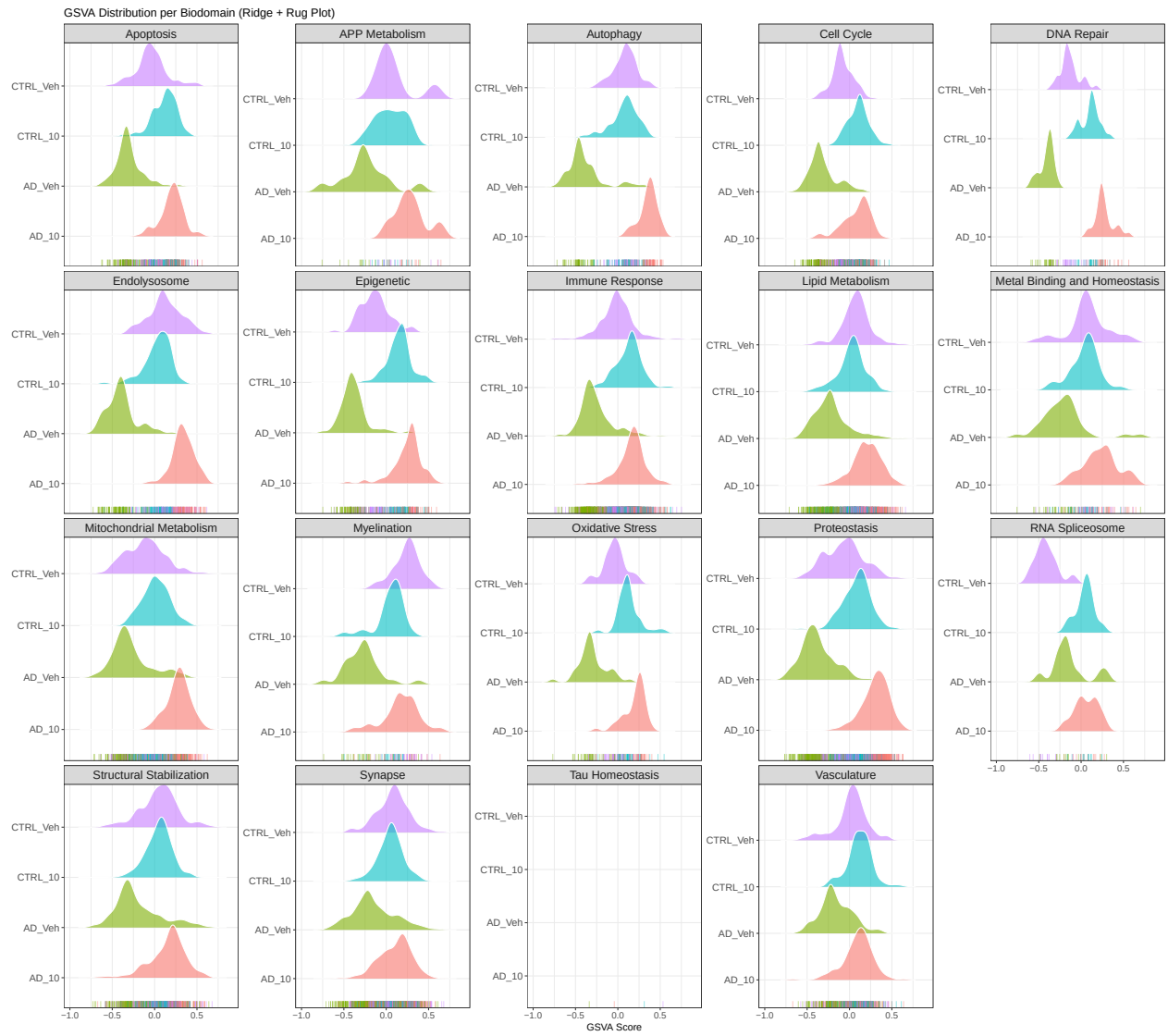
(1) Ridge plot for GSVA score



(2) Raincloud plot for GSVA score



(3) Rug and density plot for GSVA score



```

# Session information

## R version 4.4.0 (2024-04-24)
## Platform: aarch64-apple-darwin20
## Running under: macOS Sonoma 14.3.1
##
## Matrix products: default
## BLAS:   /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRblas.0.dylib
## LAPACK: /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/lib/libRlapack.dylib; LAPACK v
##
## locale:
## [1] en_US.UTF-8/en_US.UTF-8/en_US.UTF-8/C/en_US.UTF-8/en_US.UTF-8
##
## time zone: America/New_York
## tzcode source: internal
##
## attached base packages:
## [1] parallel stats4 stats graphics grDevices utils datasets
## [8] methods base
##
## other attached packages:
## [1] ggdist_3.3.2 ggribes_0.5.6
## [3] GSEABase_1.66.0 graph_1.82.0
## [5] annotate_1.82.0 XML_3.99-0.18
## [7] extrafont_0.19 ggsignif_0.6.4
## [9] patchwork_1.3.0 decoupleR_2.10.0
## [11] GSVA_1.52.3 BiocParallel_1.38.0
## [13] edgeR_4.2.2 limma_3.60.6
## [15] GenomicFeatures_1.56.0 biomaRt_2.60.1
## [17] gprofiler2_0.2.3 RColorBrewer_1.1-3
## [19] data.table_1.16.4 org.Hs.eg.db_3.19.1
## [21] AnnotationDbi_1.66.0 clusterProfiler_4.12.6
## [23] ggfortify_0.4.17 pheatmap_1.0.12
## [25] EnhancedVolcano_1.22.0 ggrepel_0.9.6
## [27] apeglm_1.26.1 DESeq2_1.44.0
## [29] SummarizedExperiment_1.34.0 Biobase_2.64.0
## [31] MatrixGenerics_1.16.0 matrixStats_1.5.0
## [33] reshape2_1.4.4 Matrix_1.7-2
## [35] Signac_1.14.0 Seurat_5.2.1
## [37] SeuratObject_5.0.2 sp_2.2-0
## [39] rtracklayer_1.64.0 GenomicRanges_1.56.2
## [41] GenomeInfoDb_1.40.1 IRanges_2.38.1
## [43] S4Vectors_0.42.1 BiocGenerics_0.50.0
## [45] knitr_1.49 lubridate_1.9.4
## [47] forcats_1.0.0 stringr_1.5.1
## [49] dplyr_1.1.4 purrr_1.0.4
## [51] readr_2.1.5 tidyr_1.3.1
## [53] tibble_3.2.1 ggplot2_3.5.1
## [55] tidyverse_2.0.0
##
## loaded via a namespace (and not attached):
## [1] SpatialExperiment_1.14.0 R.methodsS3_1.8.2
## [3] progress_1.2.3 goftest_1.2-3
## [5] HDF5Array_1.32.1 Biostrings_2.72.1

```

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##	[13]	magick_2.8.5	MASS_7.3-64
##	[15]	httpuv_1.6.15	qvalue_2.36.0
##	[17]	withr_3.0.2	xfun_0.51
##	[19]	ggfun_0.1.8	survival_3.8-3
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##	[23]	systemfonts_1.2.1	ragg_1.3.3
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